ALESSIO CURCIO

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Artificial Intelligence & Robotics Engineer | M.Sc. Student @ Sapienza University of Rome

PROFESSIONAL SUMMARY

Software Engineer and M.Sc. student in **Artificial Intelligence & Robotics (AIRO)** at **Sapienza University of Rome**, holding a Bachelor's degree in **Applied Computer Science and Artificial Intelligence (ACSAI)** graduated with **110 / 110** (taught entirely in English). Completed a research software engineering internship at **INGV** (National Institute of Geophysics and Volcanology) developing *Seismic Viewer 4D*. Scored 30 and 30 cum laude in university programming examinations (Python and Java). Technical focus on **Multi-Agent Systems**, **Local LLMs**, **Computer Vision (YOLO)**, and **Concurrent Robotic Control**. Problem-solving background across national Mathematics and Informatics Olympiads and maximum score in the TeraBebras competition.

EDUCATION

Master of Science (M.Sc.) in Artificial Intelligence and Robotics — **2024 – Present**
Sapienza University of Rome — Faculty of Information Engineering, Informatics and Statistics — *Rome, Italy*

- **Curriculum:** Autonomous Robotics, Computer Vision, Multi-Agent Systems, Reinforcement Learning, Robot Programming.

Bachelor of Science (B.Sc.) in Applied Computer Science and AI (ACSAI) — **Graduated: 110 / 110**
Sapienza University of Rome — Degree Program Taught 100% in English — *Rome, Italy*

- **Final Graduation Grade:** 110 / 110.
- **B.Sc. Thesis & Internship:** INGV (Istituto Nazionale di Geofisica e Vulcanologia) — *Seismic Viewer 4D*.
- **Honors:** Maximum grades (30 and 30 cum laude) in University Programming Examinations (Python and Java).
- **Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Software Architecture, Networks, AI, Concurrent Programming.

Scientific High School Diploma — Applied Sciences Curriculum — **Grade: 98 / 100**
Liceo Scientifico Statale 'Vito Volterra' — *Graduated 2023*

TECHNICAL SKILLS

AI & Computer Vision:	PyTorch, TensorFlow, Keras, YOLOv11, OpenCV, Local LLMs (LM Studio), Hugging Face, Qwen3-TTS, Scikit-Learn, Reinforcement Learning.
Programming Languages:	Python, Java, C, C++, Go (Golang), TypeScript, JavaScript, SQL, Bash, R.
Backend & Distributed:	Go, Docker, Docker Compose, Node.js, WebSockets, PostgreSQL, REST APIs, POSIX Multithreading.
Frontend & UI:	React, Vue.js, TailwindCSS, HTML5/CSS3, PyQt5, Tkinter, Flet.
Tools & Systems:	Git, GitHub, Linux, Tesseract OCR, Streamlit, IBM Cloud & Watson.

KEY TECHNICAL PROJECTS (GITHUB: CRUCIO104)

OpenSyntax AI Coding Workspace — *TypeScript, Multi-Agent, React, Local LLMs, Node.js*

- Workspace using **multi-agent orchestration** for software development.
- Supports dynamic scaffolding and compilation across Python, C/C++, Java, Go, and Rust.
- Hybrid support for local LLMs (via LM Studio) and cloud endpoints (OpenAI).

Local LLM Email Agent — *Python, LM Studio, Desktop GUI, Local AI Automation*

- Inbox assistant operating locally without sending email data to external cloud services.
- Uses LM Studio for message summaries, task identification, and rule-based priority tagging.

Echo Real-Time Messaging Platform — *Go (Golang), Vue.js, WebSockets, Docker Compose, PostgreSQL*

- Messaging platform built for the Web & Software Architecture examination.
- WebSocket duplex communication for 1-to-1 chats and group channels with Docker containerization.

League of Legends Autonomous Bot (YOLOv11) *Python, YOLOv11, Computer Vision, PyTorch, Screen Analysis*

- Real-time computer vision agent using **YOLOv11** for minion detection and priority scoring.
- Executes target prioritization, skillshots, and turret proximity safety routines.

Voice Cloning & Speech Synthesis (Qwen3-TTS) *TypeScript, React, Docker, Speech AI, Qwen3-TTS*

- Web client for zero-shot voice cloning and multilingual speech synthesis using Qwen3-TTS.
- Built with React and containerized with Docker.

Stock Market Predictor (LSTM) *Python, LSTM Neural Networks, Streamlit, Pandas, Yahoo Finance*

- Time-series forecasting models using Long Short-Term Memory (LSTM) recurrent neural networks.
- Yahoo Finance data integration, technical moving averages, and interactive Streamlit UI.

PDF File Extractor & OCR Desktop App *Python, PyQt5, Tesseract OCR, DOCX/TXT Export*

- Desktop tool to extract text from digital and scanned PDFs via Tesseract OCR.
- Multi-language recognition and document export in DOCX and TXT.

Concurrent & Robotic Systems Programming *C, C++, POSIX Threads, Mutex Synchronization, Kinematics*

- Multi-threaded concurrent programs with thread synchronization (mutex, semaphores).
- Robotic kinematic transformations and sensor feedback processing algorithms.

RESEARCH & INDUSTRY EXPERIENCE

Software Engineering Research Intern — Seismic Viewer 4D — INGV (Istituto Nazionale di Geofisica e Vulcanologia) ***Curricular Internship & Thesis***

- Designed and developed **Seismic Viewer**, an interactive 4D desktop application for spatio-temporal seismic data visualization.
- Engineered Python scientific backend for spatial clustering and automated fault plane estimation via PCA/RANSAC fitting (angular error < 1.0°).
- Built WebGL 3D rendering frontend with binary streaming protocol achieving an 80% data reduction vs. JSON, sustaining interactive framerates.
- Integrated real-time INGV GOSSIP API, GNSS ground uplift measurements, and 3D seismic tomography.

Data Analytics & Conversational AI — IBM & Sapienza University ***Academic Project***

- Big Data analytics for profiling models from public data streams; built customer support chatbots with IBM Cloud and Watson services.

Software & Telecom Labs — Ericsson & Fondazione Mondo Digitale ***Technical Program***

- Technical problem solving in telecommunications, applied object-oriented software design, and robotics.

COMPETITIONS, HONORS & CERTIFICATIONS

- **TeraBebras Competition:** Achieved the **maximum score** in computational thinking and algorithm design.
- **Mathematics & Computer Science Olympiads:** Qualified for regional and national stages in Italian Mathematics, Informatics and Problem Solving Olympiads.
- **Bocconi University Mathematics Games:** Advanced to national ranking rounds.
- **National Cybersecurity Competition:** Cleared qualification stages in security and cryptography challenges.
- **Languages: Italian** (Native), **English** (CEFR B2 Certification, English B.Sc. degree curriculum, Oxford study abroad).